Chapter 28 Review

Campbell Biology, AP Edition, 12e

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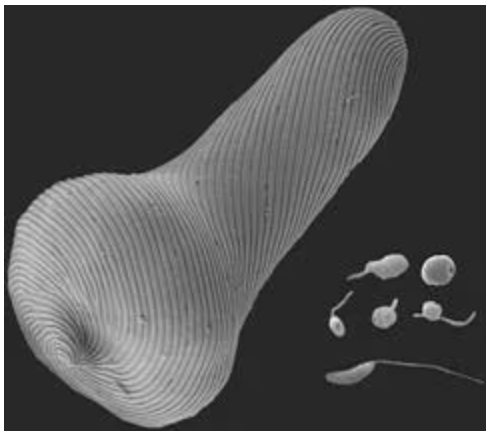

Summary of Key Concepts

Summarize the changes that occur in the cellular slime mold *Dictyostelium* when food is scarce **(BIG IDEAS 2 & 3)**, and infer why such "behavior" might be adaptive **(BIG IDEA 1)**.

Concept 28.1: Most eukaryotes are single-celled organisms

- Domain Eukarya includes many groups of ****__protists__, along with plants, animals, and fungi. Unlike prokaryotes, protists and other eukaryotes have a nucleus and other membrane-enclosed organelles, as well as a cytoskeleton that enables them to have asymmetric forms and to change shape as they feed, move, or grow.
- Protists are structurally and functionally diverse and have a wide variety of life cycles. Most are unicellular. Protists include photoautotrophs, heterotrophs, and ****__mixotrophs__.
- Current evidence indicates that eukaryotes originated by ****__endosymbiosis__ when an archaeal host (or a host closely related to the archaea) engulfed an alpha proteobacterium that would evolve into an organelle found in all eukaryotes, the mitochondrion.
- Plastids are thought to be descendants of cyanobacteria that were engulfed by early eukaryotic cells. The plastid-bearing lineage eventually evolved into ****__red algae__ and ****__green algae__. Other protist groups evolved from ****__secondary endosymbiotic__ events in which red algae or green algae were themselves engulfed.
- In one hypothesis, eukaryotes are grouped into four supergroups: ****__Excavata__, ****__SAR__, ****__Archaeplastida__, and ****__Unikonta__.

Describe similarities and differences between protists and other eukaryotes.

Key Concept/Eukaryote Supergroup	Major Groups	Key Morphological Characteristics	Specific Examples
CONCEPT 28.2 **Excavates include protists with modified mitochondria and protists with unique flagella** ? *What evidence indicates that the excavates form a clade?*	**Diplomonads and parabasalids**	Modified mitochondria	*Giardia, Trichomonas*
	Euglenozoans Kinetoplastids Euglenids	Spiral or crystalline rod inside flagella	*Trypanosoma, Euglena*
CONCEPT 28.3 **SAR is a highly diverse group of protists defined by DNA similarities** ? *Although they are not photosynthetic, apicomplexan parasites such as* Plasmodium *have modified plastids. Describe a current hypothesis that explains this observation.*	**Stramenopiles** Diatoms Oomycetes Brown algae	Hairy and smooth flagella	*Phytophthora, Laminaria*
	Alveolates Dinoflagellates Apicomplexans Ciliates	Membrane-enclosed sacs (alveoli) beneath plasma membrane	*Pfiesteria, Plasmodium, Paramecium*
	Rhizarians Radiolarians Forams Cercozoans	Amoebas with threadlike pseudopodia	*Globigerina*
CONCEPT 28.4 **Red algae and green algae are the closest relatives of plants** ? *On what basis do systematists place plants in the same supergroup (Archaeplastida) as red and green algae?*	**Red algae**	Phycoerythrin (photosynthetic pigment)	*Porphyridium, Palmaria*
	Green algae	Plant-type chloroplasts	*Chlamydomonas, Ulva*
	Plants	(See Chapters 29 and 30.)	Mosses, ferns, conifers, flowering plants
CONCEPT 28.5 **Unikonts include protists that are closely related to fungi and animals** ? *Describe a key feature for each of the main protist subgroups of Unikonta.*	**Amoebozoans** Tubulinids Slime molds Entamoebas	Amoebas with lobe-shaped or tube-shaped pseudopodia	*Amoeba, Dictyostelium*
	Opisthokonts	(Highly variable; see Chapters 31–34.)	Choanoflagellates, nucleariids, animals, fungi

Key Concept/Eukaryote Supergroup	Major Groups	Key Morphological Characteristics	Specific Examples
CONCEPT 28.2 Excavates include protists with modified mitochondria and protists with unique flagella ? What evidence indicates that the excavates form a clade?	Diplomonads and parabasalids Euglenozoans Kinetoplastids Euglenids	Modified mitochondria Spiral or crystalline rod inside flagella	Giardia, Trichomonas Trypanosoma, Euglena
CONCEPT 28.3 S A R is a highly diverse group of protists defined by DNA similarities ? Although they are not photosynthetic, apicomplexan parasites such as Plasmodium have modified plastids. Describe a current hypothesis that explains this observation.	Stramenophiles Diatoms Oomycetes Brown algae Alveolates Dinoflagellates Apicomplexans Ciliates Rhizarians Radiolarians Forams Cercozoans	Hairy and smooth flagella Membrane-enclosed sacs (alveoli) beneath plasma membrane Amoebas with threadlike pseudopodia	Phytophthora, Laminaria Pfiesteria, Plasmodium, Paramecium Globigerina
CONCEPT 28.4 Red algae and green algae are the closest relatives of plants ? On what basis do systematists place plants in the same supergroup (Archaeplastida) as red and green algae?	Red Algae Green Algae Plants	Phycoerythrin (photosynthetic pigment) Plant-type chloroplasts (See Chapters 29 and 30.)	Porphyridium, Palmaria Chlamydomonas, Ulva Mosses, ferns, conifers, flowering plants
CONCEPT 28.5 Unikonts include protists that are closely related to fungi and animals ? Describe a key feature for each of the main protist subgroups of Unikonta.	Amoebozoans Tubulinids Slime molds Entamoebas Opisthokonts	Amoebas with lobe-shaped or tube-shaped pseudopodia (Highly variable; see Chapters 31-34.)	Amoeba, Dictyostelium Choanoflagellates, nucleariids, animals, fungi

Concept 28.6: Protists play key roles in ecological communities

- Protists form mutualistic and parasitic relationships that affect their symbiotic partners and many other members of the community.
- Photosynthetic protists are among the most important ****__producers__ in aquatic communities. Because they are at the base of the food web, factors that affect photosynthetic protists affect many other species in the community.

Describe several protists that are ecologically important.

*For suggested answer, see**Appendix A.*